

Numerical Linear Algebra First-Year Exam, January 2019

Do 4 (four) problems.

1. Let $A = U\Sigma V^*$ be the singular value decomposition of $A \in \mathbb{C}^{m \times n}$ with $\text{rank}(A) = p$ and $p \leq n \leq m$.
 - (a) Show $\text{Col}(A) = \text{Span}\{u_1, u_2, \dots, u_p\}$, where $u_1, \dots, u_p$ are the first p columns of U .
 - (b) Show $\text{Null}(A^*) = \text{Span}\{u_{p+1}, u_{p+2}, \dots, u_m\}$.
 - (c) Show that A^*A is invertible if and only if A is full rank.
2. Let matrix $A \in \mathbb{C}^{m \times n}$, with $n < m$. Let vector $b \in \mathbb{C}^m$, and let r denote the residual vector $r = b - Ax$.
 - (a) Show that x solves the least-squares problem $\min \|b - Ax\|_2$ if and only if $r \in \text{Null}(A^*)$.
 - (b) Suppose A is full rank, and describe how to find the least-squares solution using the QR decomposition of A .
3. This problem has three parts.
 - (a) Show that if $A \in \mathbb{C}^{m \times m}$, A has a Schur decomposition.
 - (b) Show that if $T \in \mathbb{C}^{m \times m}$ is normal and triangular, then T is diagonal.
 - (c) Show that if $A \in \mathbb{C}^{m \times m}$ is normal and λ is an eigenvalue of A , then the geometric multiplicity of λ is equal to the algebraic multiplicity of λ .
4. Let $\|\cdot\|$ be a subordinate (induced) matrix norm. If A is $n \times n$ invertible and E is $n \times n$ with $\|A^{-1}\| \|E\| < 1$, then show
 - (a) $A + E$ is nonsingular.
 - (b)

$$\|(A + E)^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\| \|E\|}.$$

5. Consider the matrix A given by

$$A = \begin{pmatrix} 1 & 1 & 2 & -3 \\ 1 & 10 & -1 & 1 \\ 2 & -1 & 13 & -2 \\ -3 & 1 & -2 & 25 \end{pmatrix}.$$

- (a) Show that A is positive definite.
- (b) What can you say about the location of each of the eigenvalues of A ? Your answer should be in the form of an interval or a union of intervals.
- (c) Suppose the eigenvalues of A are all distinct (they are) and satisfy $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4$. Describe an algorithm that could be assured to converge to λ_4 .